



MANIPAL SCHOOL OF INFORMATION SCIENCES

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Master of Engineering – M.E in Computer Science and Engineering

Course Plan

Course Name	:	Applied Machine Learning
Course Code	:	AME 5002
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Dr. Arockiaraj
Name of the Program Coordinator	:	Dr. Sathyendranath Malli

Date:	13-07-2026
Signature of Program Coordinator	
with Date	

Date:	10-07-2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering – M.E in Computer Science and Engineering, programme are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.
PEO 2	Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering – M.E in Computer Science and Engineering, graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop software based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Applied Machine Learning [AME 5002]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Data science fundamentals
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Apply different types of supervised machine learning algorithms to practical problems.	12	PO3	3
CO2	Apply different types of unsupervised machine learning algorithms to practical problems.	12	PO4	3
CO3	Analyse different types of machine learning paradigms.	12	PO1	4



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1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	1 (15 days)	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Applying	Applying and analysing	Applying and analysing
Pattern	Answer all 5 questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks.	Applying and analyzing machine learning models	Answer all 10 full questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks.
Schedule	As per academic calendar.	22-Oct-26	As per academic calendar.
Topics covered	Introduction to Machine Learning-Linear regression-Logistic regression.	Apply and analyse the concepts of supervised and unsupervised machine learning algorithms	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Overview of Supervised (regression and classification) with practical examples	C01
L2	Overview of unsupervised (clustering and dimensionality reduction) with examples	C01
L3	Overview of semi-supervised, and reinforcement learning with example	C01
L4	Machine learning nomenclature: raw data, types of features and outputs, feature vector	C01
L5	Handling missing data, encoding categorical variables, feature scaling and normalization	C01
L6	Classification and regression using decision trees	C01
L7	Decision tree - splitting criteria	C01
L8	Decision tree - entropy	C01
L9	Decision tree - information gain	C01
L10	Decision tree - Gini impurity	C01
L11	Overfitting in decision trees	C01



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L12	Pruning in decision trees	C01
L13	Linear regression model	C02
L14	Linear regression -metrics (MSE, RMSE, R-sqaure)	C02
L15	Linear regression - estimation, and interpretation of coefficients	C02
L16	Introduction to bias/variance trade-off	C02
L17	Regularization methods (Ridge and Lasso)	C02
L18	Logistic regression for binary classification	C02
L19	Sigmoid function, decision boundary, model interpretation	C02
L20	Performance metrics such as accuracy, precision, recall, F1-score, ROC curve and AUC.	C02
L21	Cross-validation techniques	C02
L22	Nearest neighbours' algorithms	C02
L23	Probabilistic modelling of data using parameters-Introduction to maximum likelihood estimation (MLE) of parameters	C02
L24	Naive Bayes model for classification	C02
L25	Ensemble learning methods - bagging and boosting techniques	C03



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L26	Ensemble learning methods - Random Forest	C03
L27	Introduction to hyperparameter tuning and model improvement	C03
L28	K-means clustering - computing distances and similarities, limitations	C03
L29	K-means clustering - computing distances and similarities, limitations	C03
L30	Hierarchical clustering - dendrogram construction	C03
L31	Types of linkage	C03
L32	Dimension reduction using principal component analysis (PCA)-concept	C03
L33	PCA-mathematics and applications	C03
L34	Gaussian Mixture Models (GMM): probabilistic clustering	C03
L35	Mixture of Gaussian distributions	C03
L36	Expectation-maximization (EM) algorithm for parameter estimation	C03



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1.5 References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at <https://www.statlearning.com/>
2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at <https://www.manning.com/books/grokking-machine-learning>
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III – Online resource available at <http://ciml.info/>
5. Module: Introduction to Machine Learning (<https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html>).
6. Learning path: Understand data science for machine learning (<https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/>).

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



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1.7 Course Timetable

I Semester						Room: LG01-LH08			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday									
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									

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1.8 Assessment Plan

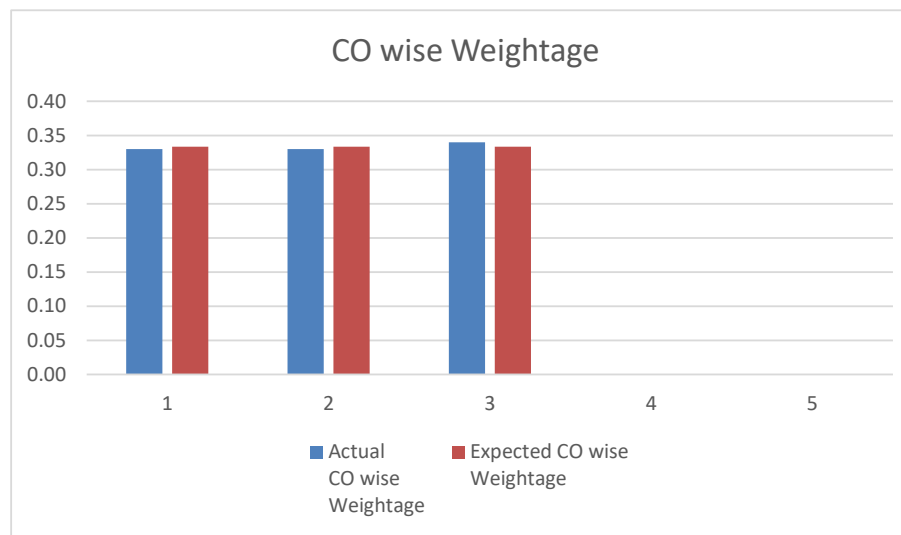
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Apply different types of supervised machine learning algorithms to practical problems.	30		30	0.33	0.33
C02	Apply different types of unsupervised machine learning algorithms to practical problems.	20	6	30	0.33	0.33
C03	Analyse different types of machine learning paradigms.		14	40	0.34	0.33
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1}/1.0 + \text{ESE marks for CO1} / 2)/100$$



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	- Performance is measured using MSE attainment level. - Reference: question paper and answer scheme. - MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	1	- Performance is measured using assignments/quiz attainment level. - Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	- Performance is measured using ESE attainment level. - Reference: question paper and answer scheme. - ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2				Y	
CO3	Y				
Average Articulation Level	*		*	*	